



Overview of German Power Market

With focus on onshore wind capture prices and investment implications

Prepared by

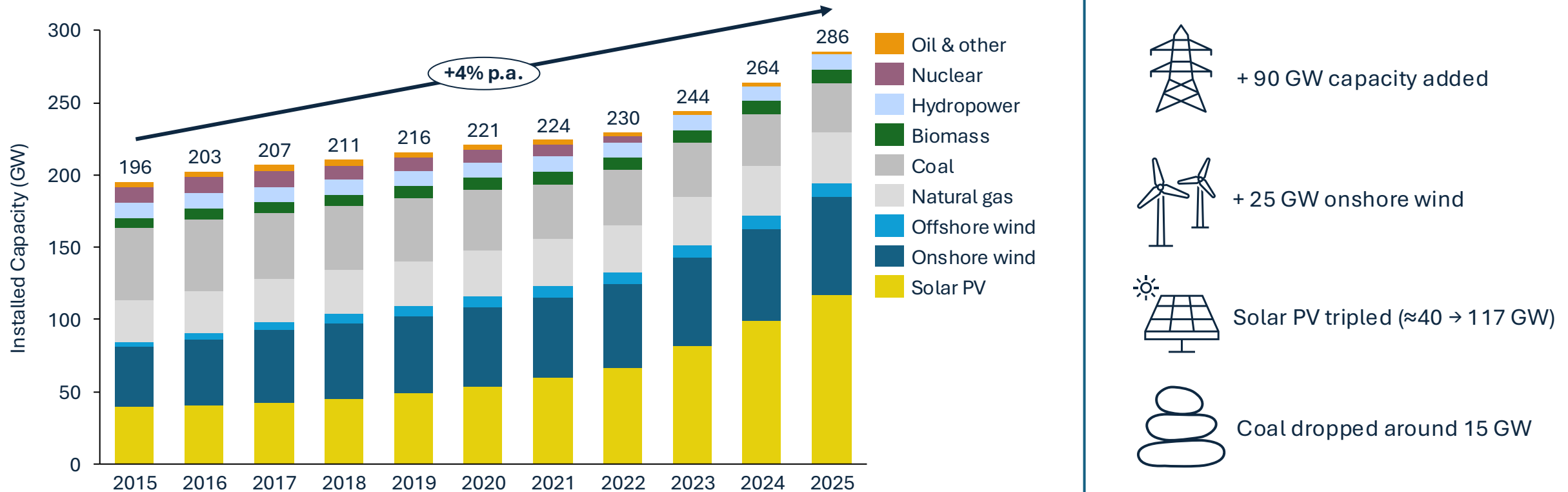
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Agenda

- Overview of the German power market
- Capture Price development
- Investment insights
- Conclusion and future work

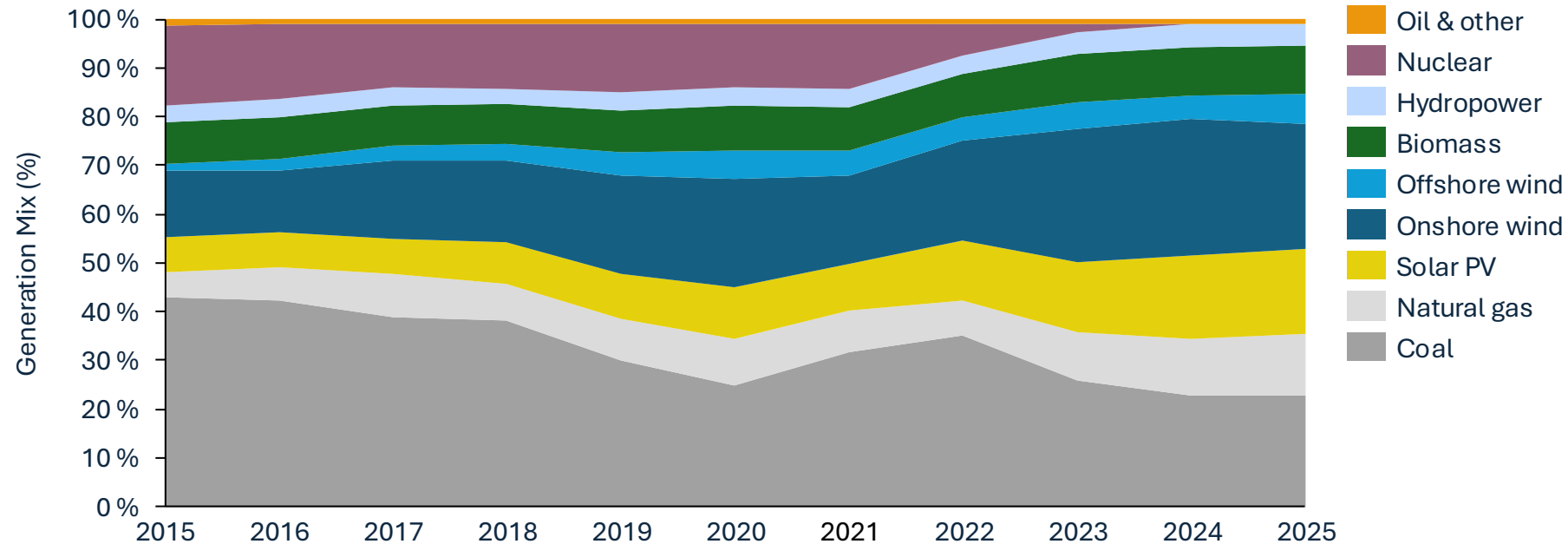


German Installed capacity grew 4 % p.a. since 2015



Source: Case study data supplemented with Fraunhofer ISE energy database

Renewables make up around 60 % German power generation mix in 2025

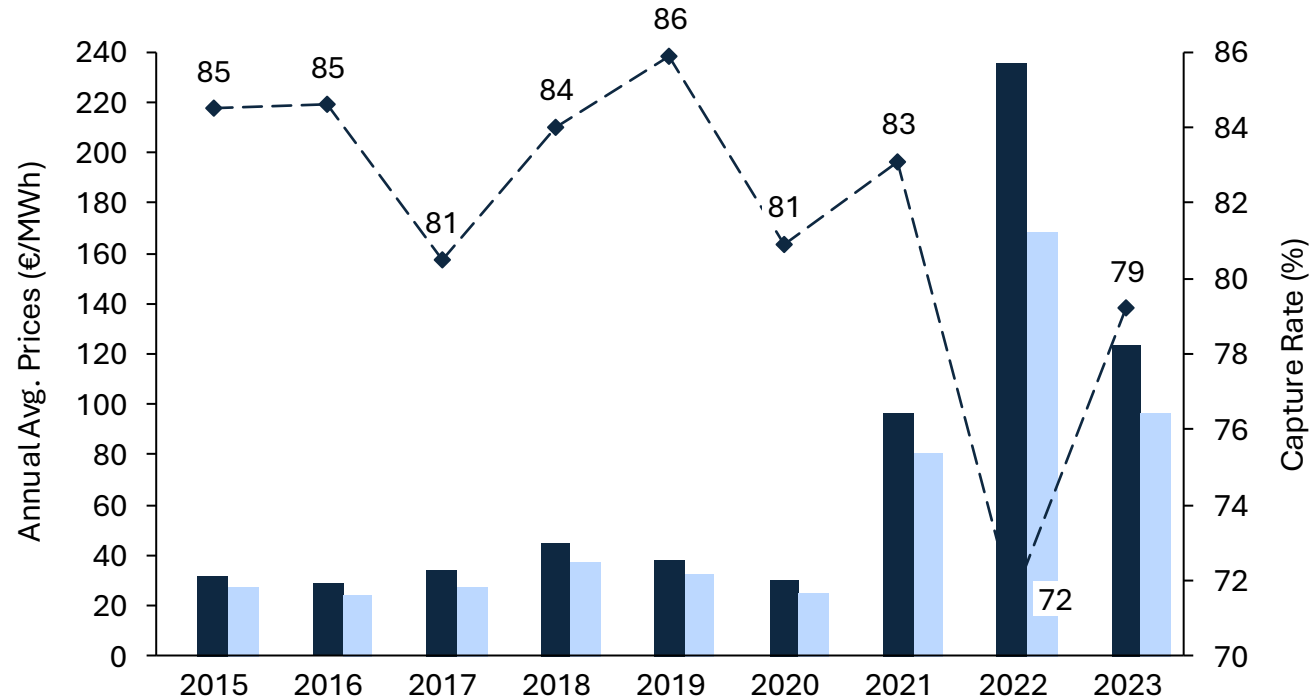


Source: Case study data supplemented with Fraunhofer ISE energy database

- Onshore wind alone contributes ~25% in 2025
- Solar around 17 % in 2025
- Nuclear phase out 2023 – 2024
- Coal and gas dominate baseload however in decline

Onshore wind capture price highly impacted by high gas prices

■ Baseload price (€/MWh) ■ Onshore wind capture price (€/MWh) —◆— Onshore wind capture rate (%)



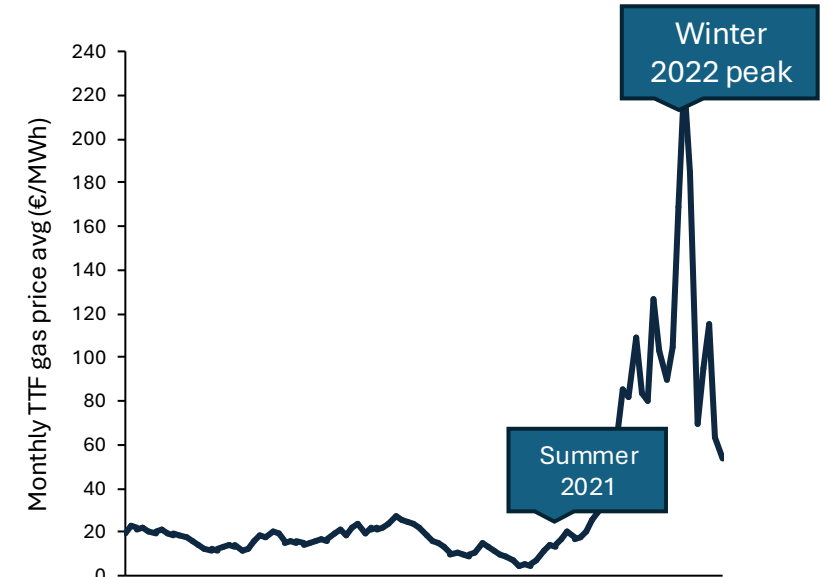
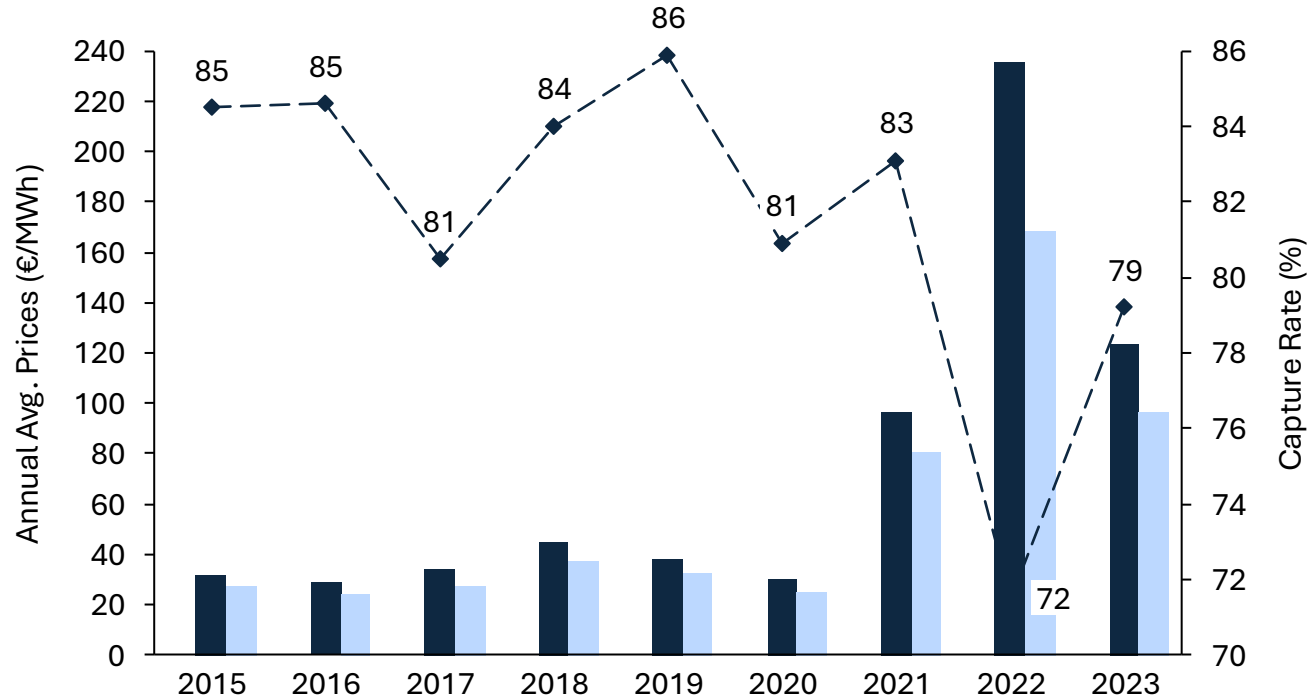
$$\text{Capture price} = \frac{\text{Avg. revenue (€)}}{\text{Avg. production (MWh)}}$$

$$\text{Capture rate (\%)} = \frac{\text{Capture price}}{\text{Baseload market price}}$$

- Capture price and baseload price ‘almost’ track 1:1
- Gas crisis summer 2021 – winter 2022 caused a deep gap
- Onshore wind capture rate dropped **around 10 %** amid high gas prices in 2022

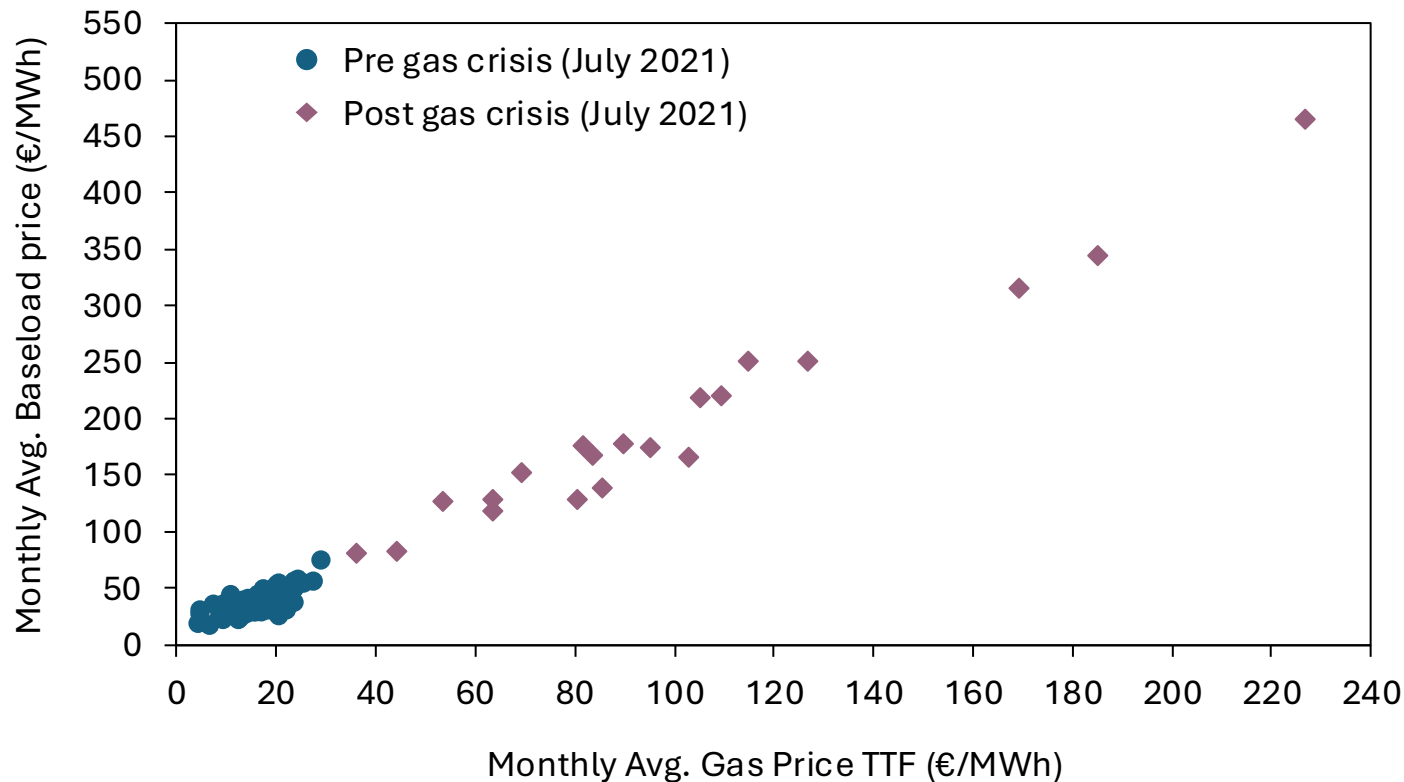
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Baseload prices highly correlated with gas prices especially during crises



Low gas price regime (round blue) – baseload price and gas price relation: 1.24 slope and R^2 0.42

High gas price regime (diamond red) – baseload price and gas price relation: 1.94 slope and R^2 0.96

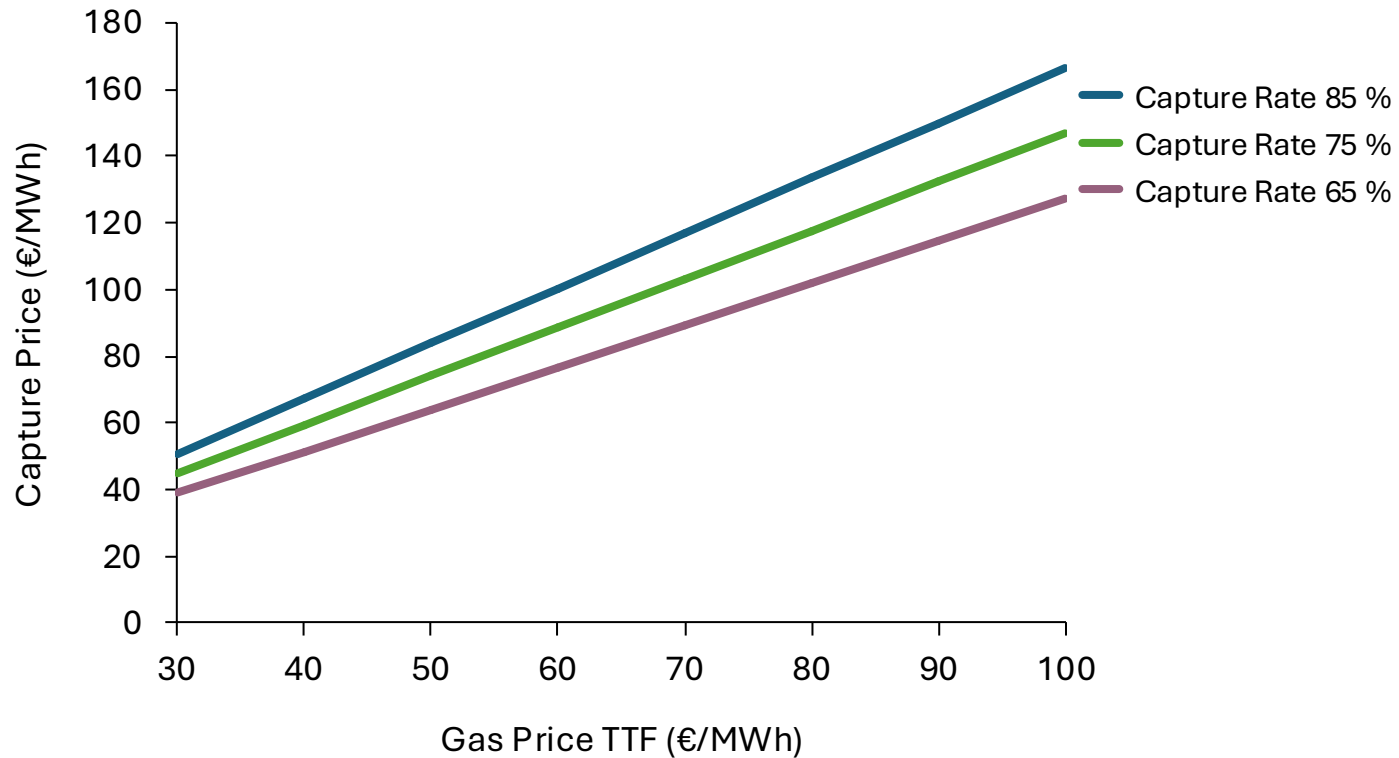
Pre gas crisis:

1 €/MWh gas price ↑ = 1.24 €/MWh baseload price ↑

Post gas crisis:

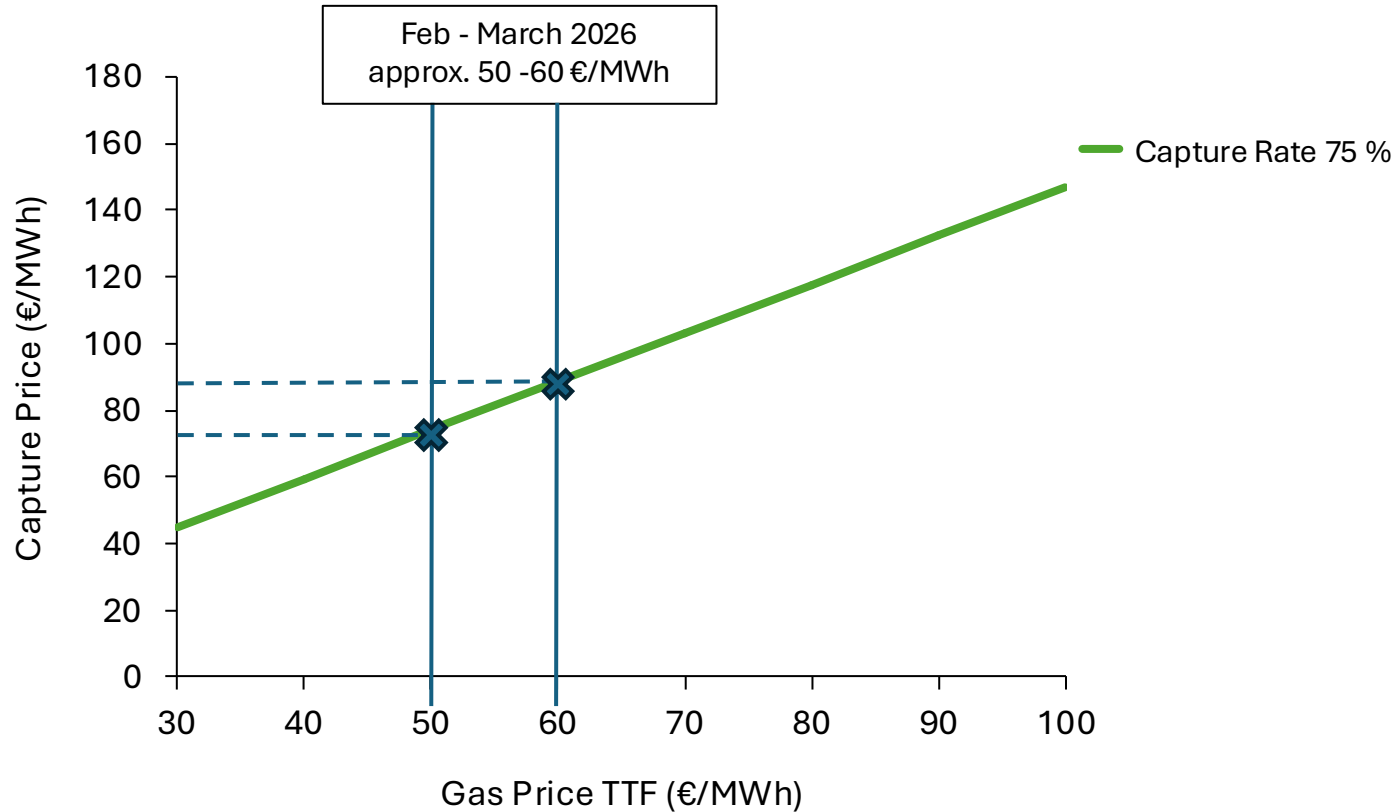
1 €/MWh gas price ↑ = 1.94 €/MWh baseload price ↑

PPA pricing benchmark - high gas price regime



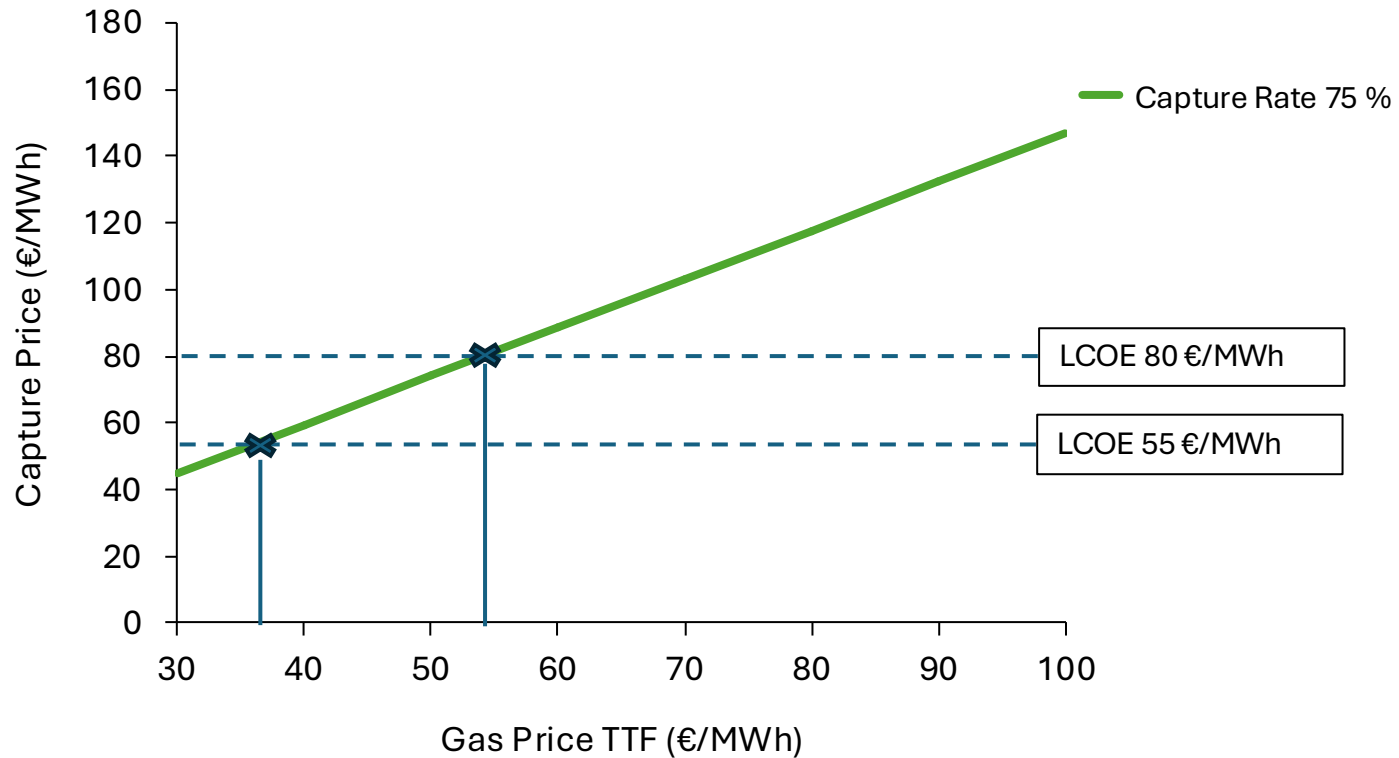
- Gas prices drive capture prices helping benchmark PPA floor price
- Higher the capture rate, higher the impact of gas prices

PPA pricing benchmark - high gas price regime



- Current forwards geopolitically driven – gas shortages next 2-3 years

Higher gas prices help onshore wind breakeven faster



- TTF breakeven price is 35 – 55 €/MWh depending on the LCOE
- Below breakeven point subsidies needed i.e. EEG

Conclusion

- RE gen share in the German market is on the rise (60 % in 2025)
- Gas prices have a very high impact on onshore wind prices
- PPA floor prices need to take into account TTF gas prices
- Hybrid configurations BESS with onshore wind and Solar PV should be considered for stable capture prices

